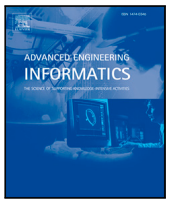
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Full length article



# Bayesian cooperative probabilistic Transformer for remaining useful life prediction with uncertainty estimation in industrial equipment

Shushuai Xie [a,b,d], Wei Cheng [a,b], Zelin Nie [a,b,*], Ji Xing [c], Xuefeng Chen [a,b], Qian Huang [c], Rongyong Zhang [c], Yunjie Yang [d]

[a] The National Key Laboratory of Aerospace Power System and Plasma Technology, Xi'an Jiaotong University, Xi'an 710049, PR China
[b] School of Mechanical Engineering, Xi'an Jiaotong University, Xi'an 710049, PR China
[c] China Nuclear Power Engineering Co. Ltd., Beijing, PR China
[d] Institute for Imaging, Data and Communications, School of Engineering, The University of Edinburgh, EH9 3FG Edinburgh, UK

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ABSTRACT

Remaining useful life (RUL) prediction is a key task in prognostics and health management (PHM) of industrial equipment, crucial for enabling reliable predictive maintenance and enhancing equipment reliability. Transformer is widely used for RUL prediction due to its ability to capture remote dynamics and nonlinear responses. However, mainstream Transformers focus on overall prediction accuracy and lack the capability to effectively assess uncertainty of RUL prediction. Furthermore, equal treatment of multi-source information or incorporating additional multi-source information fusion modules during Transformer training can increase prediction uncertainty. To address these challenges, a novel Bayesian cooperative probabilistic Transformer method for machine RUL prediction is proposed, which deeply combines the Bayesian network with the Transformer to comprehensively estimate prediction uncertainty. Prediction uncertainty in Transformer consists of attentional uncertainty, cognitive uncertainty, and aleatoric uncertainty. The core idea is that a negative log-likelihood loss function is first designed to capture attentional uncertainty, while cognitive and aleatoric uncertainties are quantified using Bayesian approach and noise channel. Additionally, an uncertainty-guided multi-source fusion strategy dynamically integrates low and high-uncertainty models for accurate RUL prediction. The performance and advantages of this method are verified by experiments on the C-MAPSS dataset and the nuclear circulating water pump (NCWP) dataset.

## 1. Introduction

At present, accurate and reliable RUL prediction is used as an important basis for maintenance plan customization in PHM, and its accurate and reliable prediction results are beneficial for changing the maintenance plan from scheduled maintenance to predictive maintenance [1–4]. As a result, predictive maintenance plans based on RUL predictions improve the highly reliable continuous uptime of industrial equipment, increasing the operational safety and economic efficiency of the business.

Currently, existing RUL prediction methods include data-driven, model-based, and hybrid-based approaches [5–7]. The latter two methods require physical or mathematical statistical modeling based on coupling mechanisms in mechanical systems to simulate mechanical degradation trends [3,8,9]. In fact, the difficulty in modeling complex mechanical systems, the need for expert experience in model simplification, and model parameter estimation have greatly limited model-based and hybrid methods for mechanical RUL prediction.

Deep Learning (DL) has been significantly used in the field of RUL prediction for industrial equipment because of architectural flexibility and superior nonlinear responsiveness [10,11]. For example, Jin et al. [12] constructed the adaptive dynamic with variable convolutional neural network (CNN) to realize RUL prediction for machinery. Zhou et al. [13] constructed a two-threaded gated recurrent neural network (RNN) for gear RUL prediction. However, the CNN is not good at capturing long distance time series features. The RNN acquires global information through sequential order and cannot realize parallel computation [14]. In recent years, since the Transformer with the attention mechanism can effectively avoid the above problems, the Transformer for RUL prediction is gradually coming into the view of researchers [15,16]. For example, Zhang et al. [17] constructed a pruned adaptive optimal lightweight Transformer for predicting RUL. Zou et al. [18] designed the depth-based multiscale window Transformer to predict the bearing RUL.

* Corresponding author.
*E-mail address:* niezelin@xjtu.edu.cn (Z. Nie).

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However, traditional DL focuses on the overall prediction accuracy without being able to effectively quantify the uncertainty of RUL predictions, which is clearly detrimental to maintenance decision-making. In general, the uncertainty of DL-based RUL prediction is categorized into cognitive uncertainty and aleatoric uncertainty [19–21]. Specifically, cognitive uncertainty is mainly due to specific network parameters, while aleatoric uncertainty is mainly due to environmental noise. The Bayesian neural network (BNN) is widely used for uncertainty modeling by transforming the network parameters into normal distributions [22–24]. For example, Peng et al. [25] used the depth combination of Bayesian network and RNN to predict RUL. Lin et al. [26] constructed the combined uncertainty quantization and Bayesian deep learning framework for RUL prediction. Zhang et al. [27] constructed the hybrid Bayesian network for RUL prediction with uncertainty.

Existing researchers have also made preliminary attempts to combine different uncertainty estimation methods with Transformer for RUL prediction to capture model uncertainty. For example, Peng et al. [28] constructed the convolutional Bayesian Transformer for bearing RUL prediction and uncertainty quantification. Peng et al. [29] proposed a Bayesian sparse self-attention fusion of multi-scale sensing information with Bayesian backpropagation for uncertainty information. Cheng et al. [30] proposed a Bayesian adversarial Transformer model for long-term RUL prediction, where adversarial learning was used to mitigate the cumulative variance loss of the prediction and Bayesian networks were used to quantify the uncertainty of the RUL prediction. Hu et al. [31] quantified the uncertainty in the RUL prediction using the quantile regression method to avoid the transition confidence induced by the traditional transformer prediction. The existing Transformer structure quantifies the uncertainty of RUL mainly through the acquired cognitive uncertainty and aleatoric uncertainty, completely ignoring the uncertainty introduced by the self-attention mechanism in the Transformer structure. Self-attention mechanism as the core part of the Transformer, using the attention obtained by point product is difficult to input the dependencies and uncertainties in the sequence data, which may lead to the inability to accurately estimate the attentional uncertainty of the RUL. Due to the special nature of the calculation of the attentional mechanism, it is not possible to estimate attentional uncertainty using the BNN method, and a special method for estimating attentional uncertainty needs to be designed. Therefore, attentional uncertainty, cognitive uncertainty and aleatoric uncertainty are all important for the uncertainty estimation of Transformer-based RUL prediction. Unfortunately, existing studies have not effectively integrated these uncertainty factors to comprehensively assess the uncertainty of RUL predictions.

In addition, the Transformer is more focused on capturing time-series contextual information, and using a single signal or treating multi-source sensor information equally during training does not seem to be a reasonable choice. Degradation data acquired by different sensors reflect differently the state of health of industrial equipment at different stages of operation, and noisy information from insensitive degradation information may make prediction results more uncertain. To address this problem, existing researchers have also used different approaches to solve the problem of multi-source information fusion in Transformer models, for example, Zhang et al. [32] embedded the trend enhancement module and the practice feature attention module into the traditional Transformer model for multi-sensor information fusion. Wang et al. [33] constructed a two-view Transformer for RUL prediction that fuses temporal and spatial information from timestamps and sensor nodes by cascading temporal and spatial views. Zhu et al. [34] proposed a parallel multi-scale convolutional neural network and Transformer model to achieve simultaneous extraction of different time-invariant and sensor features in a time series and RUL prediction using feature fusion. Hu et al. [35] integrated the long short-term memory (LSTM) and CNN into Transformer to extract global and local features and complete multi-source information fusion for RUL prediction. However, these mainstream information fusion methods are mainly realized by adding an additional fusion module to the Transformer, which leads to a more complex network structure and increases the uncertainty of the RUL prediction. Therefore, how to minimize the prediction uncertainty of the Transformer model while achieving effective fusion of multi-source sensor information is an issue worthy of further research.

To address the above problems, we propose a new Bayesian collaborative probabilistic Transformer for RUL prediction with uncertainty estimation in industrial equipment. This method combines Bayesian with Transformer to comprehensively quantify the uncertainty of RUL through attentional uncertainty, cognitive uncertainty, and aleatoric uncertainty. Specifically, we propose a specific loss function to optimize multilayer perceptron (MLP) network (i.e. probabilistic attention model, PAM) to capture attentional uncertainty. Cognitive uncertainty and aleatoric uncertainty due to noise are quantified by the Bayesian approach and the introduced noise channel. Furthermore, in contrast to the approach of adding a multi-source information fusion module, we design a new strategy for prediction uncertainty-guided multi-source information fusion (UGMIF). This strategy effectively avoids the prediction uncertainty associated with mainstream information fusion modules without changing the structure of the Transformer model. Specifically, two models are trained for low and high uncertainty sensor information, which are referred to as the low uncertainty model (LU-model) and the high uncertainty model (HU-model). During the prediction process, these two models dynamically interact and fuse to perform the final prediction based on predictive uncertainty. The effectiveness and advantages of the proposed method are verified by a large number of experiments. The main contributions of the proposed method are as follows:

(1) A novel Bayesian cooperative probabilistic Transformer for mechanical RUL prediction is proposed, which comprehensively and accurately assesses the uncertainty of prediction results by obtaining attentional uncertainty, cognitive uncertainty, and aleatoric uncertainty. In particular, attentional uncertainty is captured by the proposed probabilistic attention model, cognitive uncertainty is captured by the BNN, and aleatoric uncertainty is captured by the designed noise channel, and a novel paradigm is constructed for the comprehensive assessment of probabilistic Transformer-based RUL prediction uncertainty considering the three uncertainty.
(2) A novel view of multi-source information fusion strategy driven by predictive uncertainty is proposed, which effectively avoids the additional predictive uncertainty introduced by mainstream multi-source information fusion modules through the proposed fusion strategy. During the inference stage, predictive uncertainty is used to dynamically guide the fusion of the LU-model and the HU-model, enabling joint execution of the final prediction.
(3) The NCWP bearing test bench was first constructed to validate the performance of the proposed method for RUL prediction in complex mechanical systems. The effectiveness and superiority of the proposed method are validated using C-MAPSS dataset and NCWP dataset.

The rest of the paper is organized as follows. The main theoretical foundations involved of the proposed method are presented in Section 2. The proposed Bayesian cooperative probabilistic Transformer for RUL prediction method principle and implementation details are mainly explained in Section 3. The validity and advantages of the proposed method are verified by extensive experiments on two datasets in Section 4. The main summary of the full article and an overview of future research directions are given in Section 5.

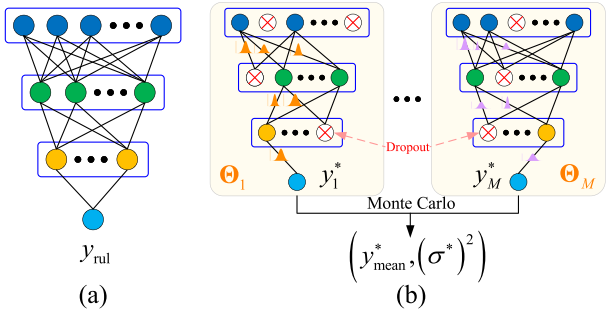



**Fig. 1.** Schematic of Bayesian network for uncertainty estimation. (a) Deterministic DL. (b) BNN based on Monte Carlo dropout.

## 2. Theoretical basis

### 2.1. Bayesian neural network

To capture the cognitive uncertainty of the RUL prediction, BNN is used for uncertainty modeling by distributing the model parameters $\boldsymbol{\Theta}$ probability distributions as shown in Fig. 1. The fixed network parameters of traditional deterministic DL result in deterministic RUL predictions. However, the parameters of the BNN network obey a probability distribution, and the Monte Carlo dropout strategy is used to obtain multiple RUL predictions in order to obtain the mean and standard deviation of the predictions, and thereby assess the uncertainty of the RUL predictions. Define $\mathcal{D} = \{\mathbf{X}, \mathbf{Y}\} = \left\{\left(\mathbf{x}_n, y_n\right)\right\}_{n=1}^{N}$ as the dataset of inputs $\mathbf{x}_n \in \mathbb{R}^{L\times C}$ with corresponding labels $y_n$, where $N$ is the sample size, $L$ is sample length, and $C$ is the multi-source sensor channel. Assuming that $\mathbf{x}^*$ is new observational information and $y^*$ is predicted RUL, the prediction uncertainty of $y^*$ is estimated as follows.

$$\mathcal{P}\left(y^* \left|\mathbf{x}^*, \mathcal{D}\right.\right) = \int \mathcal{P}\left(y^* \left|\mathbf{x}^*, \boldsymbol{\Theta}\right.\right) \cdot \mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right) d\boldsymbol{\Theta} \tag{1}$$

where $\mathcal{P}\left(y^* \left|\mathbf{x}^*, \boldsymbol{\Theta}\right.\right)$ is the likelihood estimate of the model for $\mathbf{x}^*$ and $y^*$, and $\mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right)$ is the posterior probability of the model. However, the $\mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right)$ is difficult to compute, and variational inference is often used to solve this problem [22]. The easy-to-compute $\mathcal{Q}\left(\boldsymbol{\Theta}\right)$ is used to approximate the $\mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right)$, and minimizing the distance between two distributions using Kullback–Leibler (KL) dispersion to obtain a $\mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right)$ approximation.

$$KL\left(\mathcal{Q}\left(\boldsymbol{\Theta}\right) \| \mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right)\right) = \int \mathcal{Q}\left(\boldsymbol{\Theta}\right) \log \frac{\mathcal{Q}\left(\boldsymbol{\Theta}\right)}{\mathcal{P}\left(\boldsymbol{\Theta} \left|\mathcal{D}\right.\right)} d\boldsymbol{\Theta} \tag{2}$$

Minimizing Eq. (2) to obtain the best $\mathcal{Q}\left(\boldsymbol{\Theta}\right)$, the approximate distribution of Eq. (1) can be expressed as:

$$\mathcal{P}\left(y^* \left|\mathbf{x}^*, \mathcal{D}\right.\right) \approx \int \mathcal{P}\left(y^* \left|\mathbf{x}^*, \boldsymbol{\Theta}\right.\right) \cdot \mathcal{Q}\left(\boldsymbol{\Theta}\right) d\boldsymbol{\Theta} \tag{3}$$

### 2.2. Deterministic transformer

The deterministic Transformer is widely used in language translation because its attention mechanism can focus on the contextual properties of long sequence information [36,37]. The proposed method uses the encoding part of the traditional Transformer as the main framework. Given a sequence of marker inputs, the positional encoding is calculated.

$$\begin{cases} pe_{(pos,2i)} = \sin\left(pos/10000^{2i/d_m}\right) \\ pe_{(pos,2i+1)} = \cos\left(pos/10000^{2i/d_m}\right) \end{cases} \tag{4}$$

where $pos$ is the actual location of the sample, $d_m$ is the sample dimension, and $i$ is the sample index. The marked information is then projectively transformed into queries ($Q$), keys ($K$), and values ($V$).

$$Q = W_Q\mathbf{x}, K = W_K\mathbf{x}, V = W_V\mathbf{x} \tag{5}$$

where $W_*$ represents a projection parameter matrix. Further, contextual information is captured by the attention mechanism.

$$\text{Atten}(Q, K, V) = \text{softmax}(\frac{QK^{\text{T}}}{\sqrt{d_k}})V \tag{6}$$

where $d_k$ is the dimension of $Q$. To obtain spatial positional relationships at different locations, multi-head attention in the Transformer is employed.

$$\text{MHA}(Q, K, V) = \text{Concat}\left(\text{h}_1, \text{h}_2, \ldots, \text{h}_H\right) W^O \tag{7}$$

where $\text{h}_i = \text{Atten}\left(Q_i, K_i, V_i\right), i = 1, 2, \ldots, H$, $W^O$ is trained to ensure consistency between input and output dimensions, $\text{Concat}\left(\cdot\right)$ represents the splicing operation, and $H$ represents the number of attention head.

In addition, the features of the multi-head attention output are fed into the feed-forward network to obtain decision information for use in the downstream processing.

$$\text{FFN}\left(x\right) = \text{ReLU}\left(W_1\tilde{\mathbf{z}} + b_1\right) W_2 + b_2 \tag{8}$$

where $\tilde{\mathbf{z}}$ is the feature vector, $W$ is the weight, and $b$ is the bias.

## 3. Proposed method

The general architecture of the proposed Bayesian collaborative probabilistic Transformer method for RUL prediction in industrial equipment in Fig. 2. The architecture is divided into three main parts: data acquisition, Bayesian cooperative probabilistic Transformer and RUL prediction based on LU-model and HU-model. More details of the data acquisition can be found in the experiment analysis. Details of the other two parts are as follows.

### 3.1. Probabilistic attention

For traditional Transformer, $Q$ and $K$ are usually used to obtain the attention of the context. Based on Eq. (6) we can obtain the attention weight by two observations, which is calculated as follows:

$$\alpha_{ij} = \frac{Q_i \cdot K_j}{\sqrt{d_k}} \tag{9}$$

where $i, j \in \{1, 2, \ldots, L\}$. Then, the output value of the $i$th $x_i$ by weighted attention is calculated as follows:

$$\tilde{x}_i = \sum_{j=1}^{L} \frac{\alpha_{ij}}{\sum_{l=1}^{L} \alpha_{il}} V_j \tag{10}$$

The above equation can capture dependencies between input sequences. The degradation processes of complex mechanical systems are typically stochastic and variable. The deterministic attention calculated by Eq. (9) is ineffective in capturing noise-induced attentional errors, thereby increasing uncertainty in prediction results. Therefore, the pointwise attention computation in the traditional Transformer faces two problems: (1) It is sensitive to input perturbations and feature redundancy, thereby lacking robustness. (2) All attention weights are deterministic point estimates and fail to express uncertainty, limiting the credibility and generalization of the model. To address these issues, we introduce a probabilistic attention model (PAM), as shown in Fig. 3. We assume $\alpha_{ij} \sim \mathcal{N}\left(\mu_{ij}, \sigma_{ij}^2\right)$, where the $u_{ij}$ and $\sigma_{ij}$ are computed via the MLP, i.e., $\mu_{ij}, \sigma_{ij} = \text{MLP}\left(Q_i, K_j\right)$. For updating the probability parameters $\mu_{ij}$ and $u_{ij}$ during network forward propagation, we employ a reparameterization strategy [38,39].

$$\tilde{\alpha}_{ij} = \mu_{ij} + \tau\sigma_{ij}, \tau \sim \mathcal{N}\left(0, 1\right) \tag{11}$$

where $\tau$ is sampled from a normal distribution during the experiment. This reparameterization strategy follows the standard reparameterization mechanism, ensuring gradient propagation during sampling and improving model stability. We replace the deterministic attention weights in Eq. (10) with the probabilistic attention weights $\tilde{\alpha}_{ij}$ after

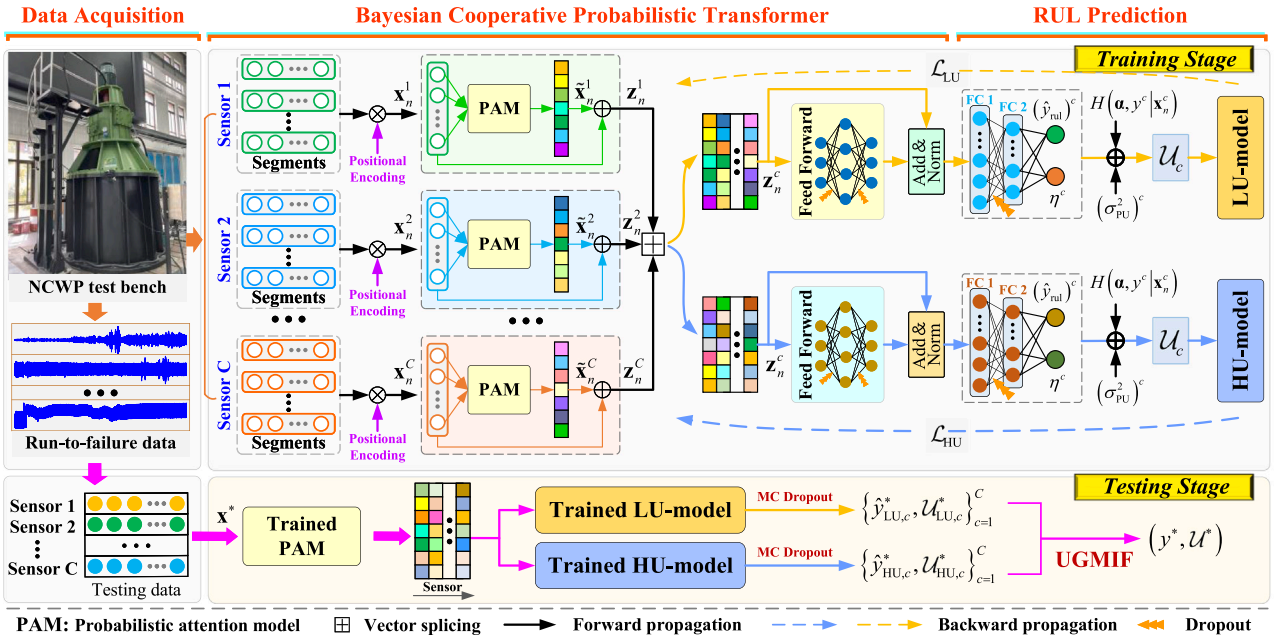



**Fig. 2.** Structure illustration of the proposed Bayesian cooperative probabilistic Transformer.

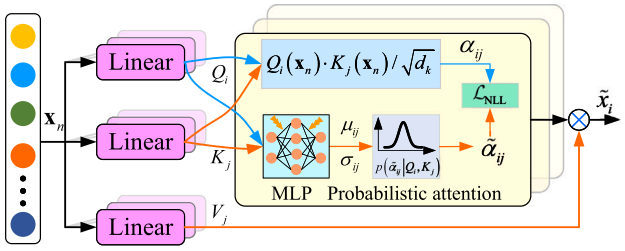



**Fig. 3.** Diagram of probabilistic attention model (PAM).

training completion. In addition, $\tilde{\alpha}_{ij}$ is trained using Eq. (20). From the perspective of information entropy analysis, the proposed PAM mechanism introduces explicit distribution modeling into the mapping of $Q_i$ and $K_i$ representations. This approach effectively compresses input information redundancy while retaining key temporal dependencies, thereby enhancing model generalization.

### 3.2. Uncertainty modeling

We comprehensively analyze the sources of uncertainty in Transformer-based RUL predictions for industrial equipment, which are mainly categorized into attentional uncertainty, cognitive uncertainty and aleatoric uncertainty. The detailed description is developed below.

#### 3.2.1. Attentional uncertainty

Attentional uncertainty primarily reflects uncertainty in model dynamics regarding its attention to different input features or regions. This type of uncertainty reflects differences in the model's perception of the importance of different input features and is directly related to the model's attention mechanism. Similar to calculating cognitive uncertainty, during the test period, we generate $\boldsymbol{\alpha}$ with the number $M$. We estimate attentional uncertainty through information entropy.

$$H\left(\boldsymbol{\alpha}, y^* \left| \mathbf{x}^* \right.\right) = \mathcal{H}\left(\frac{1}{M}\sum_{m=1}^{M} \mathcal{P}\left(y^* \left| \mathbf{x}^*, \boldsymbol{\alpha}_m \right.\right)\right) - \frac{1}{M}\sum_{m=1}^{M} \mathcal{H}\left(\mathcal{P}\left(y^* \left| \mathbf{x}^*, \boldsymbol{\alpha}_m \right.\right)\right) \tag{12}$$

where $\boldsymbol{\alpha} = \left\{\alpha_{ij} \left| i, j \in \{1, \ldots, L\} \right.\right\}$, $\mathcal{H}(\cdot)$ represents an information entropy function.

#### 3.2.2. Cognitive uncertainty

Cognitive uncertainty primarily arises from imperfect knowledge due to model structure, parameter selection, and limited training data. Essentially, it results from generalization errors caused by inadequate learning of the data distribution, which can be mitigated by optimizing the model or augmenting the dataset. Cognitive uncertainty is usually obtained by the BNN. BNN can employ various probability distribution assumptions to model and quantify prediction uncertainty. The parameters of BNN are typically assumed to follow Gaussian, Laplace, or Student-t distributions. Although Laplace and Student-t distributions offer advantages in handling heavy-tailed data and outliers, the Gaussian distribution provides mathematical tractability, training stability, and good generalizability for RUL prediction. Currently, mainstream BNNs still predominantly use the Gaussian distribution as the predictive assumption to effectively quantify uncertainty in machine RUL prediction. Actually, learning about posterior distribution $\mathcal{P}(\boldsymbol{\Theta} \left| D \right.)$ is a tricky thing. According to [27,40], we define $\mathcal{Q}(\boldsymbol{\Theta})$ approximations instead of the $\mathcal{P}(\boldsymbol{\Theta} \left| D \right.)$. We define the $s$th weight matrix $\mathbf{W}_s \in \mathbb{R}^{R \times C}$ of the model parameter distribution $\mathcal{Q}(\boldsymbol{\Theta})$ as a mixture distribution with two Gaussians, then

$$\begin{cases} \mathcal{Q}\left(\mathbf{W}_s\right) = \prod_{r=1}^{R} \mathcal{Q}\left(\mathbf{w}_r\right) \\ \mathcal{Q}\left(\mathbf{w}_r\right) = \mathcal{P}_s \mathcal{N}\left(\mathbf{m}_r, \sigma^2 \mathbf{I}_C\right) + \left(1 - \mathcal{P}_s\right) \mathcal{N}\left(0, \sigma^2 \mathbf{I}_C\right) \end{cases} \tag{13}$$

where $\mathcal{P}_s \in [0, 1]$, $\sigma$ is set to $10^{-23}$ [27], $\mathbf{m}_r \in \mathbb{R}^C$ is a variational parameter, and $\mathbf{I}_C$ is the identity matrix. Similarly, the bias $\mathbf{b}_s$ is distributed using a Gaussian approximation:

$$\mathcal{Q}\left(\mathbf{b}_s\right) = \mathcal{N}\left(\mathbf{m}, \sigma^2 \mathbf{I}_R\right) \tag{14}$$

where $\mathbf{m}$ is the variational parameter. Minimizing the KL divergence can be translated into maximizing the logarithmic evidence lower bound (ELBO) direct reference in Python.

$$KL\left(\mathcal{Q}(\boldsymbol{\Theta}) \| \mathcal{P}(\boldsymbol{\Theta} \left| D \right.)\right) = \int \mathcal{Q}(\boldsymbol{\Theta}) \log \mathcal{P}\left(\mathbf{Y} \left| \mathbf{X}, \boldsymbol{\Theta} \right.\right) d\boldsymbol{\Theta} - \text{ELBO} \tag{15}$$

Based on the MC-dropout unbiased estimation of the first term of Eq. (15):

$$\begin{aligned}&\int \mathcal{Q}(\boldsymbol{\Theta}) \log \mathcal{P}(\mathbf{Y}|\mathbf{X}, \boldsymbol{\Theta}) d\boldsymbol{\Theta} \\ &= \sum_{m=1}^{M} \int \mathcal{Q}(\boldsymbol{\Theta}) \log \mathcal{P}\left(\mathbf{y}_m|\mathbf{x}_m, \boldsymbol{\Theta}\right) d\boldsymbol{\Theta}\end{aligned} \tag{16}$$

where $M$ is the number sampled by MC-dropout. Therefore, our log-likelihood for the predicted probability in Eq. (1) can be approximated by the MC-dropout.

$$\begin{aligned}\log \mathcal{P}(y^*|\mathbf{x}^*, \mathcal{D}) &= \log \int \mathcal{P}(y^*|\mathbf{x}^*, \boldsymbol{\Theta}) \cdot \mathcal{P}(\boldsymbol{\Theta}|\mathcal{D}) d\boldsymbol{\Theta} \\ &\approx \log \int \mathcal{P}(y^*|\mathbf{x}^*, \boldsymbol{\Theta}) \cdot \mathcal{Q}(\boldsymbol{\Theta}) d\boldsymbol{\Theta} \\ &\approx \log \left( \frac{1}{M} \sum_{m=1}^{M} \mathcal{P}\left(y^*|\mathbf{x}^*, \boldsymbol{\Theta}_m\right) \right)\end{aligned} \tag{17}$$

Therefore, for test data $\mathbf{x}^*$ cognitive uncertainty is calculated.

$$\sigma_{\mathrm{PU}}^2 = \frac{1}{M} \sum_{m=1}^{M} \left( f_{\hat{\boldsymbol{\Theta}}_m}\left(\mathbf{x}_m^*\right) - E\left(f_{\hat{\boldsymbol{\Theta}}_m}\left(\mathbf{x}_m^*\right)\right) \right)^2 \tag{18}$$

where $\hat{\boldsymbol{\Theta}}_m \sim \mathcal{Q}\left(\hat{\boldsymbol{\Theta}}\right)$ is the parameters of the network that completed the training.

### 3.2.3. Aleatoric uncertainty

Aleatoric uncertainty originates from inherent randomness or irreducible noise within the data, reflecting constraints due to data quality or sensor measurement accuracy. It cannot be eliminated solely by increasing model complexity or dataset size but requires finer probabilistic modeling for quantification (e.g., modeling aleatoric uncertainty in network outputs). Unlike most existing BNN-based uncertainty modeling, they fix or ignore the aleatoric uncertainty variance $\eta^2$ caused by noise. To capture aleatoric uncertainty, we add a noise output channel to capture aleatoric uncertainty. It is worth noting that aleatoric uncertainty $\eta^2$ is bounded via a loss function.

### 3.2.4. Total uncertainty

In fact, under low-noise conditions, sensor data exhibit less intrinsic randomness and lower aleatoric uncertainty. Clear and unambiguous signal characteristics enable the model to more effectively learn data feature representations, thereby reducing cognitive uncertainty. In this scenario, prediction uncertainty is dominated by attentional uncertainty, primarily characterized by ambiguity regarding the model's determination of feature importance. Conversely, in high-noise environments, the random variability of data increases, significantly amplifying both aleatoric and cognitive uncertainties. Consequently, the model's attentional mechanism struggles to identify clear feature patterns, tending instead toward dispersion. Therefore, the dynamic interaction among these three types of uncertainty indicates that jointly modeling them is essential for dynamically adapting to various noise environments, thereby ensuring robust RUL predictions in practical applications.

Based on the above uncertainty analysis, it can be concluded that the total uncertainty of the proposed method is the sum of attentional, cognitive, and aleatoric uncertainty.

$$\mathcal{U} = \sigma_{\mathrm{PU}}^2 + H\left(\boldsymbol{\alpha}, y^*|\mathbf{x}^*\right) + \frac{1}{M} \sum_{m=1}^{M} \eta^2 \tag{19}$$

## 3.3. Overall loss function

To efficiently obtain probabilistic attention distributions, we use the negative log-likelihood (NLL) loss to train MLP to obtain probabilistic attention uncertainty. The main idea is that we use MLP to learn the distribution of probabilistic attention $\tilde{\alpha}_{ij}$ and constrain probabilistic attention with deterministic attention in Eq. (9), as shown in Fig. 3. The NLL loss is calculated.

$$\begin{aligned}\mathcal{L}_{\mathrm{NLL}} = \sum_{n=1}^{N} \sum_{i,j} &\log \frac{1}{\sqrt{2\pi\sigma_{ij}^2(\mathbf{x}_n)}} \\ &\times \exp\left( -\frac{\left(\tilde{\alpha}_{ij}(\mathbf{x}_n) - Q_i(\mathbf{x}_n) \cdot K_j(\mathbf{x}_n)/\sqrt{d_k}\right)^2}{2\sigma_{ij}^2(\mathbf{x}_n)} \right)\end{aligned} \tag{20}$$

where $N$ is the number of samples, $\tilde{\alpha}_{ij}$ and $\sigma_{i,j}$ are obtained from Eq. (11), and the NLL loss is minimized along with other losses during training to complete model training.

In this article, our work aims to minimize regression loss and NLL loss, and to capture aleatoric uncertainty. Furthermore, inspired by [41,42], we add the $l_2$-regularization of $\eta^2$ to the loss term to avoid searching for unconstrained $\eta^2$. Therefore, the overall loss function of the proposed method is shown below.

$$\begin{aligned}\mathcal{L}_{\mathrm{net}} = &\frac{1}{N} \sum_{n=1}^{N} \left( \frac{1}{2\eta_n^2} (y_n - \hat{y}_n)^2 + \frac{1}{2} \log \eta_n^2 \right) \\ &+ \gamma \mathcal{L}_{\mathrm{NLL}} + \lambda \frac{1}{N} \sum_{n=1}^{N} \left( \left\| \eta_n^2 \right\|_2 \right)\end{aligned} \tag{21}$$

where $y_n$ and $\hat{y}_n$ are the true and predicted RUL, respectively, and $\gamma$ and $\lambda$ are the weight hyperparameters.

Considering the total network uncertainty, we introduce UGMIF strategy for model training. As described earlier, different sensors acquire degradation information at different stages. To better extract the degradation information of each sensor, we propose the LU-model and the HU-model, focusing on low and high uncertainty sensor information, respectively. For the LU model, more attention is given to sensor monitoring information with lower uncertainty by Eq. (22). Similarly, the HU model focuses more on higher uncertainty information by Eq. (23). It is worth noting that we use the monitoring data from each channel to train the LU-model and the HU-model using $\mathcal{L}_{\mathrm{LU}}$ and $\mathcal{L}_{\mathrm{HU}}$ losses, respectively. The two models are then combined to cover all sensor channel information using total uncertainty. In other words, we train the low uncertainty model and the high uncertainty model separately through all the channel data, and pay balanced attention to the degradation data acquired by all the sensors, not just focusing on the sensor data with obvious degradation characteristics and ignoring the sensor data with less obvious degradation. Therefore, the loss functions of the two models are calculated.

$$\mathcal{L}_{\mathrm{LU}} = \sum_{c=1}^{C} \left( 1 - \frac{\exp\left(\mathcal{U}_c\right)}{\sum_{c=1}^{C} \exp\left(\mathcal{U}_c\right)} \right) \cdot \mathcal{L}_{\mathrm{net}}^c \tag{22}$$

$$\mathcal{L}_{\mathrm{HU}} = \sum_{c=1}^{C} \left( 1 + \frac{\exp\left(\mathcal{U}_c\right)}{\sum_{c=1}^{C} \exp\left(\mathcal{U}_c\right)} \right) \cdot \mathcal{L}_{\mathrm{net}}^c \tag{23}$$

where $\mathcal{U}_c$ is the total uncertainty of the $c$th channel sample and $C$ is the sensor channel.

## 3.4. Prediction inference

Indeed, when insufficient training data or skewed distributions hinder accurate feature recognition, increased cognitive uncertainty results in greater ambiguity within the attentional mechanism. Meanwhile, the balance between attentional uncertainty and aleatoric uncertainty depends on the signal-to-noise ratio of the data. Higher noise levels increase attentional uncertainty, complicating the model's ability to focus on key features, whereas lower noise conditions have the opposite effect. Additionally, high data randomness complicates model training, further increasing cognitive uncertainty. Therefore, to effectively integrate intrinsic relationships among these uncertainties and improve model prediction performance, we propose an uncertainty-guided multi-source information fusion strategy. This strategy explicitly characterizes the interactions among the three uncertainties, enabling

the model to adaptively quantify the overall uncertainty under varying operating environments. Specifically, we dynamically combine the LU-model and HU-model using UGMIF to effectively extract distinct degradation information from multi-source sensors. The prediction for the $c$th sensor is computed as a weighted sum of outputs from the two models.

$$y_c^* = \left(1-\omega_c\right)\hat{y}_{\mathrm{LU},c}^* + \omega_c \hat{y}_{\mathrm{HU},c}^* \tag{24}$$

where $\omega_c = \exp\left(\mathcal{U}_{\mathrm{LU}}^c\right)/\left(\exp\left(\mathcal{U}_{\mathrm{LU}}^c\right)+\exp\left(\mathcal{U}_{\mathrm{HU}}^c\right)\right)$, $\mathcal{U}_{\mathrm{LU}}^c$ and $\mathcal{U}_{\mathrm{HU}}^c$ are the uncertainty of the $c$th sensor data predicted by the LU-model and the HU-model, respectively. This exponential form of weight calculation ensures that the weights are sensitive to changes in uncertainty. When uncertainty increases significantly, the sensor's weight rapidly decreases, ensuring higher fusion weights for high-quality data and theoretically reducing the overall prediction uncertainty. $\hat{y}_{\mathrm{LU},c}^* = \frac{1}{M}\sum_{m=1}^{M} y_{\mathrm{LU},c;m}^*$, $\hat{y}_{\mathrm{HU},c}^* = \frac{1}{M}\sum_{m=1}^{M} y_{\mathrm{HU},c;m}^*$, where $y_{\mathrm{LU},c;m}^*$ and $y_{\mathrm{HU},c;m}^*$ represent low and high uncertainty predictions, respectively. The proposed UGMIF is essentially a Bayesian-based fusion method, in which predictions from each sensor are treated as uncertain random variables. Therefore, the final prediction can be viewed as a weighted expectation of stochastic predictor variables guided by their respective uncertainties. This theoretically ensures that the fused result achieves minimized uncertainty in a statistical sense. Therefore, for testing data $\mathbf{x}^*$, the final RUL prediction is as follows.

$$y^* = \mathbb{E}_{y^*\sim\mathcal{U}^*}\left(y^*\right) = \sum_{c=1}^{C}\frac{\exp\left(\mathcal{U}_c^*\right)}{\sum_{c=1}^{C}\exp\left(\mathcal{U}_c^*\right)}\cdot y_c^* \tag{25}$$

where $\mathcal{U}_c^* = \left(1-\omega_c\right)\mathcal{U}_{\mathrm{LU}}^c + \omega_c \mathcal{U}_{\mathrm{HU}}^c$. To ensure uniform scaling of uncertainty values across different channel data, normalization is applied to facilitate fair fusion of multi-source data and enhance overall prediction stability. The overall prediction uncertainty after fusion is computed through normalization.

$$\mathcal{U}^* = \sum_{c=1}^{C}\mathcal{U}_c^* \cdot \frac{\mathcal{U}_{\max}^* - \mathcal{U}_c^*}{\mathcal{U}_{\max}^* - \mathcal{U}_{\min}^*} \tag{26}$$

where $\mathcal{U}_{\max}^*$ and $\mathcal{U}_{\min}^*$ are the maximum and minimum uncertainties of $\left\{\mathcal{U}_c^*\right\}_{c=1}^{C}$. Furthermore, the proposed Bayesian cooperative probabilistic Transformer training and predictive inference process is summarized in Algorithm 1.

**Algorithm 1** : Bayesian cooperative probabilistic Transformer for RUL prediction.

**Input:** Training data $\mathcal{D} = \{\mathbf{X},\mathbf{Y}\} = \left\{\left(\mathbf{x}_n, y_n\right)\right\}_{n=1}^{N}$, evaluation data $\left\{\left(\mathbf{x}_{val}, y_{val}\right)\right\}$ in $\mathcal{D}$, testing data $\mathbf{x}^*$, and dropout rate $p$.
**Output:** Distribution of trained network parameters $q\left(\hat{\boldsymbol{\Theta}}\right)$, estimated RUL $y^*$, prediction uncertainty $\mathcal{U}^*$.

**Build:** Bayesian transformer network. Initialize network parameters $\boldsymbol{\Theta}_0 \sim q\left(\boldsymbol{\Theta}_0\right)$.
*# Training process*
**for** $e$ = 1:epoch **do**
  **for** $n = 1 : N$ **do**
    Input $\left\{\left(\mathbf{x}_n, y_n\right)\right\}$.
    **for** $c = 1 : C$ **do**
      Predict $\hat{y}_n^c$, $\log \eta_{n,c}^2$ by proposed network.
    **end for**
  **end for**
  Calculate each channel $\mathcal{L}_{\mathrm{NLL}}$ by Eq. (20).
  Calculate each channel $\mathcal{L}_{\mathrm{net}}$ by Eq. (21).
  Input evaluation data $\left\{\left(\mathbf{x}_{val}, y_{val}\right)\right\}$.
  **for** $c = 1 : C$ **do**
    Initialize network parameters $\boldsymbol{\Theta}_0 \sim q\left(\boldsymbol{\Theta}_0\right)$. # Use Dropout to sample parameter.
    Calculate $\mathcal{U}_c$ by Eq. (19).
  **end for**
  Calculate $\mathcal{L}_{\mathrm{LU}}$ by Eq. (22). Calculate $\mathcal{L}_{\mathrm{HU}}$ by Eq. (23).
  Update $\boldsymbol{\Theta}_o$ by minimizing $\mathcal{L}_{\mathrm{LU}}$ or $\mathcal{L}_{\mathrm{HU}}$.
**end for**
*# Testing process*
Input testing data $\mathbf{x}^*$.
**for** $m = 1 : M$ **do**
  $\boldsymbol{\Theta}^* \sim q\left(\boldsymbol{\Theta}^*\right)$. # Use Dropout to sample parameter.
  Predict $\hat{y}_c$ and $\log \eta_c^2$ by trained network.
**end for**
$y^*$ and $\mathcal{U}^*$ are predicted by Eqs. (24)–(26).
**return** $y^*$ and $\mathcal{U}^*$.

## 4. Experiment analysis

### 4.1. Case study 1: C-MAPSS dataset

The well-known C-MAPSS dataset is used to validate the effectiveness and advantages of the proposed method and a detailed description of the dataset can be found in [43].

#### 4.1.1. Dataset description

The dataset mainly consists of four sub-datasets under four different operating conditions. The experimental setup for the four working conditions are shown in Table 1. Different sensors in each sub-dataset are used to obtain a comprehensive picture of the health of the engines, primarily through 21 sensors such as temperature, pressure, speed, etc. It is worth noting that some sensors collect constant values of degradation data during the life cycle. Therefore, only 14 sensor data are retained by removing these sensors in the experiment, as shown in Table 2.

**Table 1**
Dataset description of the C-MAPSS.

| Dataset | FD001 | FD002 | FD003 | FD004 |
|---|---|---|---|---|
| Operating conditions | 1 | 6 | 1 | 6 |
| Fault modes | 1 | 1 | 2 | 2 |
| Training engines | 100 | 260 | 100 | 249 |
| Testing engines | 100 | 259 | 100 | 248 |

**Table 2**
Description of sensor information in the C-MAPSS.

| Sensor | Description (units) | Sensor | Description (units) |
|---|---|---|---|
| T24 | Temperature at LPC | T30 | Temperature at HPC |
| T50 | Temperature at LPT | P30 | Total pressure at HPC |
| Nf | Physical fan speed | Nc | Physical core speed |
| Ps30 | Static pressure | phi | Ratio |
| NRf | Corrected fan speed | NRc | Corrected core speed |
| BPR | Bypass Ratio | phi | Bleed Enthalpy |
| W31 | HPT coolant bleed | W32 | LPT coolant bleed |

#### 4.1.2. Evaluation metric

To evaluate the performance of different RUL prediction methods, we use the root mean square error (RMSE), score function [11,43]. These indicators are defined as follows.

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2} \tag{27}$$

$$\mathrm{Score} = \begin{cases} \sum_{i=1}^{n}\left(e^{\frac{y_i-\hat{y}_i}{13}} - 1\right) & \text{if } \hat{y}_i < y_i \\ \sum_{i=1}^{n}\left(e^{\frac{\hat{y}_i-y_i}{10}} - 1\right) & \text{if } \hat{y}_i \geq y_i \end{cases} \tag{28}$$

where $n$ is the number of test engines. In addition, to quantitatively characterize prediction uncertainty, the prediction interval coverage probability (PICP) is used to evaluate the performance of a prediction method in predicting the true value over a prediction interval (PI). In

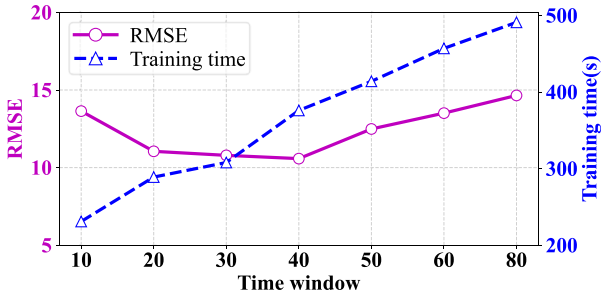



**Fig. 4.** Experimental results with different time window $l$ on C-MAPSS.

**Table 3**
Setting of network parameters for the proposed method.

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| window size | 30 | head $H$ | 5 |
| $\gamma$ | 0.2 | $\lambda$ | 0.1 |
| batch size | 256 | epoch | 200 |
| learning rate | 0.005→ 0.001 | optimizer | Adam |
| FC 1/FC 2 | 20/10 | dropout | 0.3 |

addition, PI normalized average width (PINAW) is used to assess how well predicted values fit their prediction uncertainty [44].

$$\begin{aligned} &\mathrm{PICP} = \frac{1}{N}\sum_{i=1}^{N} I_i(\beta) \\ &with \quad I_i(\beta) = \begin{cases} 1 & y_i \in \left[\hat{y}_i^{0.5-0.5\beta}, \hat{y}_i^{0.5+0.5\beta}\right] \\ 0 & otherwise \end{cases} \end{aligned} \tag{29}$$

$$\mathrm{PINAW} = \frac{1}{NI}\sum_{i=1}^{N}\left(\hat{y}_i^{0.5+0.5\beta} - \hat{y}_i^{0.5-0.5\beta}\right) \tag{30}$$

where $\beta$ is the confidence level and $I$ is the target RUL interval range. It is worth noting that the closer the PICP is to $\beta$ and the smaller the PINAW, the more reliable the RUL prediction.

### 4.1.3. Experiment details

To reduce the influence of the differences in the distribution of data from different sensors on the experiments, we used the Min–Max normalization method to preprocess the experimental data. In addition, we use the popular segmented linear function method with a RUL maximum of 125 to generate training sample labels [11,45]. The sliding time window of step size 1 and length $l$ is used to divide the dataset. Overall, a larger $l$ captures a wider range of features, but results in more model training time. In general, a larger $l$ contains more temporal information but reduces the speed of model training. Experimental results for different time windows in Fig. 4 show that the best time window is 30. For the weighting coefficients $\gamma$ and $\lambda$ in (21), the grid search is used to identify the optimal parameters. In general, both hyperparameters are searched in $\{0.01, 0.05, 0.1, 0.2, 0.3, 0.5, 0.8, 1\}$, as shown in Fig. 5. The results show that the prediction performance is better when $\gamma$ and $\lambda$ are taken as 0.2 and 0.1, respectively. The network structure for Feed Forward is 30-64-30 and the network structure for MLP is 2-10-6-2. We give more details about the network parameter settings of the proposed method in Table 3. In addition, all experiments are performed on Windows 10 system with PyTorch 1.8.1, and executed on a machine equipped with an RTX A5000 GPU and 64 GB RAM. All experiments are repeated 10 times to eliminate chance in the experimental results.

### 4.1.4. RUL prediction results and comparison

We visualize the RUL predictions and deviations over the full lifetime of the run-to-failure to capture the performance of the method in Fig. 6. The results show that the RUL predictions of the proposed

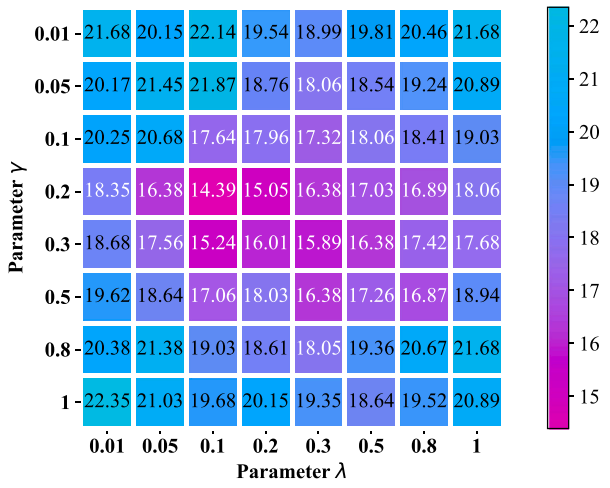



**Fig. 5.** RMSE results for weight coefficients $\gamma$ and $\lambda$ by gridded search in the C-MAPSS dataset.

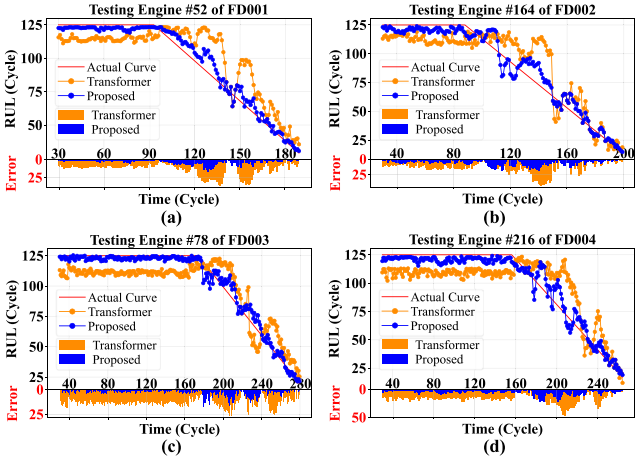



**Fig. 6.** RUL prediction and prediction errors under four sub-datasets.

method on the four sub-datasets have some bias at the early stage. However, RUL predictions become more accurate as the cycle increases, mainly due to the fact that multi-source sensors acquire more degradation information. Compared with the Transformer, the proposed method predicts more accurately, which is mainly due to the proposed UGMIF strategy that enables the model to learn the integrated degradation information to predict more accurate RUL.

We quantitatively analyze and compare the proposed method against mainstream RUL prediction approaches using RMSE and score functions. These methods include Bayesian LSTM (B-LSTM), Bayesian Transformer (B-Trans-former), gated recurrent units with dual attention mechanism (DA-GRU) [14], the basic Transformer model (Transformer) [17], Bayesian deep learning-based method (BDL-based) [25], Bayesian deep learning (BDL) [26], Bayesian mixture neural network (BMNN) [27], multi-stream attention fusion network with quantile regression (MAFN-QR) [46], Bayesian temporal convolutional Transformer (BTCT) [28], Transformer with Gaussian process (TGP) [47], hierarchical stochastic attention Transformer (HSAT) [48], and Bayesian adversarial probsparse Transformer (BAPT) [49]. The results are summarized in Table 4. Overall, the proposed method outperforms existing methods in prediction accuracy. Specifically, models with attention mechanisms generally outperform those without, indicating that attention facilitates the extraction of degradation-relevant information. Compared with multi-source fusion methods with uncertainty modeling

**Table 4**
Performance of the proposed method with other methods.

| Methods | FD001 | | FD002 | | FD003 | | FD004 | |
|---|---|---|---|---|---|---|---|---|
| | RMSE | Score | RMSE | Score | RMSE | Score | RMSE | Score |
| B-LSTM | 13.65 | 952.09 | 23.68 | 4371.66 | 17.82 | 620.38 | 25.63 | 7392.73 |
| B-Transformer | 12.37 | 245.19 | 19.85 | 1578.22 | 12.52 | 301.49 | 21.05 | 2981.27 |
| DA-GRU [14] | 11.77 | 235.81 | 16.09 | 2655.83 | 11.66 | 253.81 | 19.89 | 2499.90 |
| Transformer [17] | 12.49 | 257.71 | 21.63 | 1692.59 | 12.66 | 274.15 | 23.86 | 3163.41 |
| BDL-based [25] | 15.56 | 845.24 | 23.69 | 3245.4 | 18.37 | 458.09 | 31.75 | 7637.24 |
| BDL [26] | 18.6 | 2774.1 | 22.9 | 7743.7 | 27.9 | 19 990.6 | 28.1 | 53 295.6 |
| BMNN [27] | 14.35 | 403.28 | 18.69 | 4263.71 | 15.41 | 763.14 | 22.38 | 5894.58 |
| MAFN-QR [46] | 13.56 | 368.82 | 16.35 | 4198.66 | 14.97 | 913.06 | 19.06 | 4198.35 |
| BTCT [28] | 11.34 | 203.64 | 15.12 | 3544.15 | 11.32 | **241.15** | 19.54 | 3567.94 |
| TGP [47] | 12.09 | 302.78 | 15.58 | 2672.22 | 13.54 | 357.78 | 20.78 | 3353.47 |
| HSAT [48] | 11.35 | 297.24 | 15.29 | 1587.53 | 12.18 | 284.58 | 19.09 | 2545.11 |
| BAPT [49] | 11.12 | 214.91 | 15.04 | 1752.75 | 11.94 | 297.45 | 18.87 | 2247.69 |
| Proposed | **10.81** | **198.17** | **14.25** | **1338.91** | **11.09** | 260.27 | **17.52** | **1956.02** |

The bold entities represent the best performance.

**Table 5**
Uncertainty estimation results of RUL prediction.

| Methods | FD001 | | FD002 | | FD003 | | FD004 | |
|---|---|---|---|---|---|---|---|---|
| | PICP | PINAW | PICP | PINAW | PICP | PINAW | PICP | PINAW |
| B-LSTM | 0.922 | 0.535 | 0.815 | 0.617 | 0.903 | 0.457 | 0.807 | 0.681 |
| B-Transformer | 0.940 | 0.497 | 0.879 | 0.511 | 0.932 | 0.367 | 0.901 | 0.593 |
| BDL-based [25] | 0.938 | 0.491 | 0.876 | 0.572 | 0.924 | 0.395 | 0.845 | 0.579 |
| BDL [26] | 0.925 | 0.531 | 0.824 | 0.601 | 0.907 | 0.411 | 0.792 | 0.643 |
| BMNN [27] | 0.942 | 0.504 | 0.883 | 0.534 | 0.941 | 0.372 | 0.826 | 0.591 |
| MAFN-QR [46] | 0.943 | 0.506 | 0.897 | 0.453 | 0.929 | 0.356 | 0.892 | 0.601 |
| BTCT [28] | 0.961 | 0.487 | 0.902 | 0.479 | 0.934 | 0.318 | 0.873 | 0.584 |
| TGP [47] | 0.943 | 0.457 | 0.917 | 0.438 | 0.963 | 0.346 | 0.905 | 0.613 |
| HSAT [48] | 0.945 | 0.401 | 0.943 | 0.457 | 0.959 | 0.322 | 0.931 | 0.581 |
| BAPT [49] | 0.955 | 0.384 | 0.935 | 0.412 | 0.942 | 0.337 | 0.928 | 0.604 |
| Proposed | 0.951 | 0.321 | 0.956 | 0.369 | 0.952 | 0.315 | 0.945 | 0.562 |

(e.g., MAFN-QR), the proposed UGMIF strategy not only emphasizes sensor data with strong degradation characteristics (e.g., LU-model) but also enhances the contribution of weaker signals, enabling comprehensive degradation feature acquisition. Furthermore, unlike Transformer-based uncertainty quantification methods (e.g., BTCT, TGP, HSAT, and BAPT), the proposed approach incorporates three types of uncertainty to guide more effective multi-source information fusion. This improves the model's ability to handle complex scenarios, as reflected by its superior performance on the more difficult FD002 and FD004 sub-datasets.

### 4.1.5. Uncertainty analysis and comparison

In this study, we assume that the model parameter distributions follow a Gaussian distribution and evaluate prediction uncertainty using a 95% confidence level (CI), as shown in Table 5. Overall, prediction uncertainty differences across methods are small on FD001 and FD003, mainly due to the simplicity of their operating conditions. In contrast, the advantages of the proposed method are more pronounced on FD002 and FD004, which involve more complex operating conditions. This improvement is attributed to the newly designed UGMIF strategy, which dynamically fuses the LU-model and HU-model to enhance both prediction accuracy and stability. Additionally, the proposed method obtains a higher PINAW on the FD004 dataset, as it integrates three types of uncertainty, leading to a wider uncertainty interval. However, methods such as B-Transformer, BMNN, MAFN-QR, BTCT, TGP, and BAPT exhibit PICP values below 95%, indicating that the prediction uncertainty is underestimated.

In order to visually assess the uncertainty of the RUL predictions, we chose the 95% confidence interval to evaluate the uncertainty of the RUL prediction, resulting in Fig. 7. Note that the proposed method has a high uncertainty in the early stage, while the uncertainty continues to decrease with time, a phenomenon that is consistent with the changes

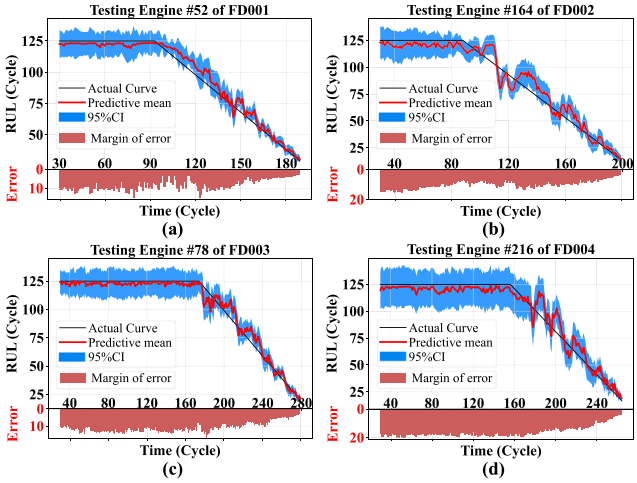



**Fig. 7.** RUL estimation results with prediction intervals for different test engines in FD001-FD004.

in Table 5. This is mainly due to the large cognitive and aleatoric uncertainties associated with less degradation information acquired by multi-source sensors in early system health states. Over time, multi-source sensors can capture more degradation information making the model cognitive uncertainty decrease. Meanwhile, the proposed UGMIF strategy prompting the fusion of two models is also a key factor in reducing the prediction uncertainty.

In addition, to visualize the change in RUL prediction uncertainty at the late stage of degradation, we demonstrate the probability density function (PDF) of the RUL of 60 prediction points at the late stage

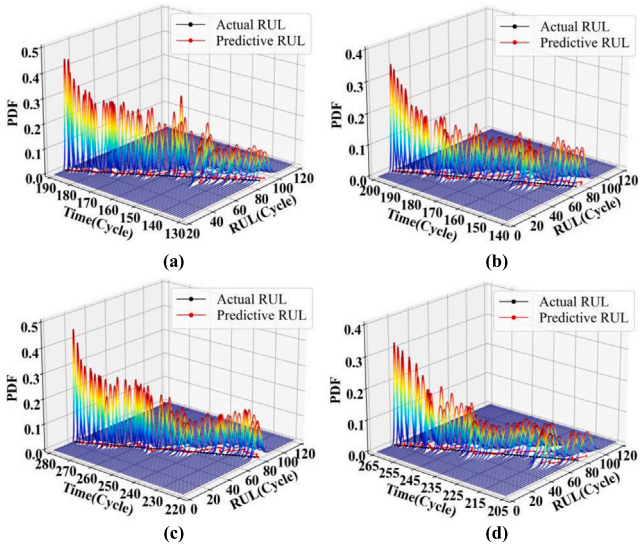



**Fig. 8.** Evaluate PDFs of RUL for different test engines in FD001-FD004. (a) #52 of FD001. (b) #64 of FD002. (c) #78 of FD003. (d) #216 of FD004.

of degradation, as shown in Fig. 8. It can be seen that almost all the PDF curves cover the actual RUL, which indicates the validity of the RUL prediction results. Furthermore, as system degradation progresses, the increasingly higher and narrower PDF curves indicate reduced prediction uncertainty, providing a reliable basis for downstream maintenance decisions.

#### 4.1.6. Ablation experiments and analysis

The effect of different modules or strategies in proposed method on the experimental results is further investigated through ablation experiments, as shown in Table 6. Note that the value corresponding to a single signal is the best value obtained from all the single signals. PICP is assessed at 95% CI. The results show that the removal of either uncertainty leads to a decrease in model performance. On the one hand, ignoring any one uncertainty factor can lead to an underestimation of prediction uncertainty (PICP below 95%), making prediction overconfident. On the other hand, the uncertainty of predictions without a comprehensive integrated assessment leads to multiple sources of information that cannot be effectively fused, which reduces the accuracy and reliability of predictions. Furthermore, considering only a single signal as a model input leads to the worst prediction accuracy, mainly because a single piece of information does not provide comprehensive and reliable feedback on the degraded state of the system. Meanwhile, without UGMIF, the prediction accuracy of the network decreases rapidly while the prediction uncertainty increases, suggesting that UGMIF is critical for RUL prediction. In fact, different sensors perceive the degradation state of the system differently at different stages, and the proposed UGMIF strategy enables multi-source sensors to feed back the degradation state of the system at different stages to improve the prediction accuracy and reduce the prediction uncertainty.

In addition, the predictive accuracy of the model in real industrial systems is limited due to insufficient acquired training data. To validate the performance and advantages of our method in the case of insufficient training data, we reduce the training data of the model from 100% to 20% and leave the testing data unchanged, resulting in Fig. 9. Compared with the deterministic Transformer, the proposed method shows better performance with limited data, which is mainly due to the UGMIF strategy that can mine more degraded information from limited data.

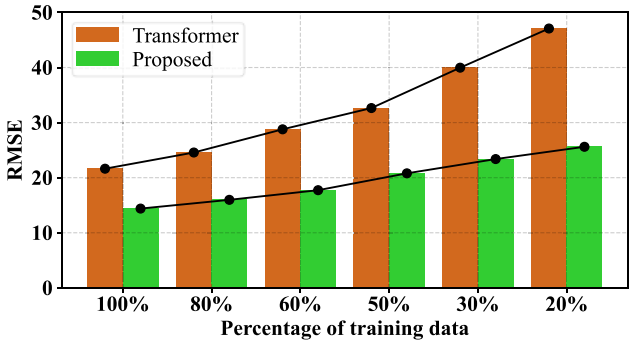



**Fig. 9.** Comparison of deterministic Transformer and the proposed method with different training data on FD002.

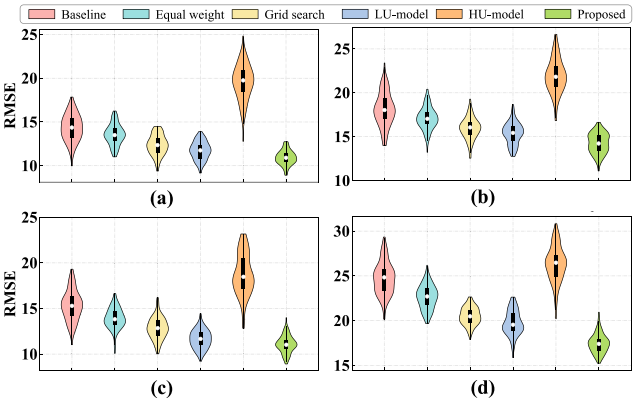



**Fig. 10.** Comparison of different multi-source information fusion strategies on the C-MAPSS dataset. (a) FD001. (b) FD002. (c) FD003. (d) FD004.

#### 4.1.7. UGMIF effectiveness analysis

To assess the effectiveness of the proposed UGMIF, we compare its performance under different information fusion strategies. The compared strategies include: treating all channels as a single input data for network training (Baseline), giving each channel data the same weight (Equal weighting), obtaining the best value of different channel weights by grid search (Grid search), focusing only on the sensor channel data that contribute high to the prediction results (LU-model), and focusing only on the sensor channel data that contribute low to the prediction results (HU-model). The experimental results are illustrated in Fig. 10, showing that the proposed method achieves superior prediction accuracy and robustness, primarily due to the following two factors: (1) Compared with common fusion strategies such as Baseline, Equal Weighting, and Grid Search, which either treat all sensor data equally and introduce noise, or focus only on channels with prominent degradation features (e.g., vibration), thereby neglecting less informative ones. The proposed UGMIF strategy not only emphasizes dominant signals (LU-model), but also leverages subtle sensor information (HU-model), leading to a more comprehensive capture of degradation characteristics and improved prediction performance. (2) Compared with the combined use of the LU-model and HU-model, focusing only on low-uncertainty channels with strong degradation features or high-uncertainty channels with weak features reduces prediction performance. This further confirms the effectiveness and robustness of the proposed UGMIF strategy.

To verify the effectiveness and advantages of the proposed multi-source information fusion strategy, we visualize the sensor channel attention weights for different engines in the test set, as shown in Fig. 11. The results show that UGMIF can assign channel-specific weights based on prediction uncertainty, which demonstrates its adaptability

**Table 6**
Results of ablation experiments in the C-MAPSS.

| Dataset | Metric | Single signal | w/o-UGMIF | w/o-AtU | w/o-CU | w/o-ArU | Proposed |
|---|---|---|---|---|---|---|---|
| FD001 | RMSE | 17.31 | 15.22 | 12.58 | 13.61 | 13.16 | 10.81 |
| | PICP | 0.871 | 0.932 | 0.944 | 0.935 | 0.937 | 0.951 |
| FD002 | RMSE | 25.19 | 18.52 | 12.37 | 15.31 | 13.12 | 14.25 |
| | PICP | 0.852 | 0.875 | 0.904 | 0.915 | 0.921 | 0.956 |
| FD003 | RMSE | 18.83 | 15.56 | 11.28 | 13.15 | 12.42 | 11.09 |
| | PICP | 0.894 | 0.925 | 0.939 | 0.931 | 0.935 | 0.952 |
| FD004 | RMSE | 29.65 | 20.35 | 14.89 | 21.36 | 15.93 | 17.52 |
| | PICP | 0.837 | 0.856 | 0.896 | 0.873 | 0.907 | 0.945 |

w/o-UGMIF: without uncertainty-guided multi-source information fusion. w/o-AtU: without attentional uncertainty. w/o-CU: without cognitive uncertainty. w/o-ArU: without aleatoric uncertainty. PICP is under 95%.

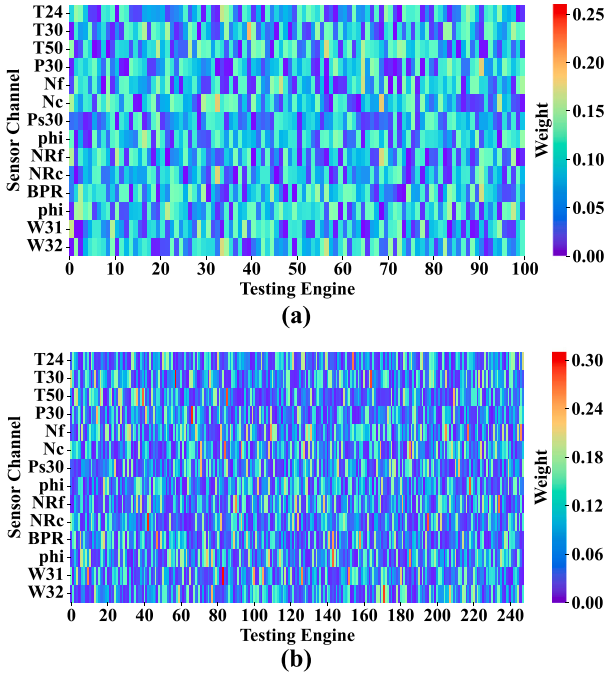



**Fig. 11.** Sensor channel weights of different engines in the test set. (a) FD001. (b) FD004.

in capturing diverse sensor degradation patterns. In particular, key channels such as P30, Nf, and T30 consistently receive high attention across multiple test cases, which aligns well with domain knowledge and enhances the interpretability of the model's decision-making process. In addition, we observe that the attention distribution varies across engine samples under different operating conditions, indicating the strong generalization and adaptability of the proposed method. In summary, the proposed UGMIF strategy can dynamically emphasize key channels and suppress redundant information, thereby increasing model robustness while maintaining fusion transparency.

#### 4.1.8. Edge deployment analysis

In resource-constrained industrial environments, the deployment of Bayesian Transformers is a practical challenge [50,51]. To further evaluate the edge applicability of the proposed method in industrial environments, we focus on the impact of the number of MC-dropout samples on the real-time inference performance within the BNN framework. Table 7 presents the effect of varying MC-dropout sample sizes on model prediction accuracy, uncertainty, and inference latency. The results indicate that using fewer MC-dropout samples accelerates inference and improves the efficiency of edge deployment. However, this may lead to RUL predictions that are highly sensitive to individual dropout realizations, resulting in higher and more volatile RMSE values. Moreover, fewer MC-dropout samples cause unstable interval estimates, an increased likelihood of missing the ground truth, and a lower PICP, indicating reduced reliability. Conversely, increasing the number of MC-dropout samples improves prediction accuracy and brings the PICP closer to the true confidence level. However, as the number of MC-dropout samples increases, the RMSE stabilizes, but this comes at the cost of higher computational overhead and increased latency. Therefore, selecting an appropriate number of MC-dropout samples is essential to balance prediction accuracy, uncertainty estimation reliability, and inference efficiency. In summary, this study sets the MC-dropout sample number to 30 on the C-MAPSS dataset, which ensures both high prediction accuracy and reliability, while meeting the real-time response requirements for edge deployment in industrial scenarios.

### 4.2. Case study 2: NCWP dataset

Furthermore, we further use the NCWP journal bearing dataset acquired in the laboratory to validate the effectiveness and scalability of the proposed method.

#### 4.2.1. Dataset description

The NCWP test bench is first built by China Nuclear Power Engineering Co., Ltd., Shanghai Apollo Machinery Co., Ltd., and Xi'an Jiaotong University to acquire full-life data from operation to failure. The test bench is built according to a certain scaling and with high fidelity, and its overall structure and components are shown in Fig. 12. It mainly consists of a motor, gearbox, bearing, shaft, impeller, pipeline, lubricating oil station, and water tank. Twelve sensors, including vibration acceleration, acoustic pressure, displacement, flow, temperature, and pressure sensors, are used to obtain comprehensive operating information of the bearings, and the sensor time-domain waveforms are shown in Fig. 13. In the experiment, we used a sampling frequency of 10240 Hz for interval sampling, the interval between two neighboring samples was 1 min, and the sampling time for each sample was 3.2 s (i.e., 32768 monitoring points were obtained for each sample). The experiment was stopped when the vibration sensor data and temperature sensor were higher than the preset thresholds and abnormal noise was monitored at the bearing, and a total of 8526 monitoring samples were obtained.

#### 4.2.2. Experiment details

We chose the observation points from 7000 to 8526 in the bearing for experimental analysis, due to the large amount of pre-existing health smoothing data has little effect on the prediction results. In addition, we use the same structural setup as in Case 1 and fine-tune the relevant parameters. The $l$ of 20 was used to divide the dataset where the training, validation, and testing sets are divided according to the ratio of 0.7, 0.1, and 0.2.

**Table 7**
The effect of MC-dropout sample size on model inference performance for the C-MAPSS dataset.

| Dataset | Metrics | MC-dropout sample size | | | | | |
|---|---|---|---|---|---|---|---|
| | | 10 | 20 | 30 | 50 | 80 | 100 |
| FD001 | RMSE | 10.98 | 10.92 | 10.81 | 10.81 | 10.80 | 10.80 |
| | PICP | 0.943 | 0.947 | 0.951 | 0.951 | 0.951 | 0.950 |
| | Latency (s) | 0.51 | 1.02 | 1.53 | 2.55 | 4.08 | 5.10 |
| FD002 | RMSE | 14.27 | 14.26 | 14.25 | 14.25 | 14.23 | 14.23 |
| | PICP | 0.934 | 0.941 | 0.956 | 0.954 | 0.953 | 0.953 |
| | Latency (s) | 0.55 | 1.10 | 1.65 | 2.75 | 4.40 | 5.50 |
| FD003 | RMSE | 11.14 | 11.12 | 11.09 | 10.09 | 10.08 | 10.08 |
| | PICP | 0.941 | 0.945 | 0.952 | 0.952 | 0.951 | 0.951 |
| | Latency (s) | 0.52 | 1.04 | 1.56 | 2.60 | 4.16 | 5.20 |
| FD004 | RMSE | 17.55 | 17.54 | 17.52 | 17.52 | 17.51 | 17.51 |
| | PICP | 0.917 | 0.931 | 0.945 | 0.946 | 0.947 | 0.947 |
| | Latency (s) | 0.57 | 1.14 | 1.71 | 2.85 | 4.56 | 5.70 |

Latency refers to the mean inference delay across the engine test set. The PICP is under 95%.

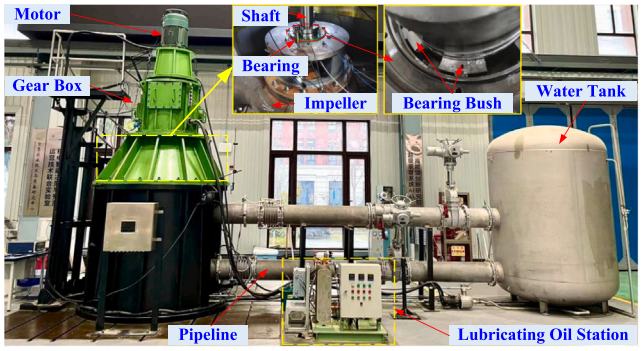



**Fig. 12.** Test bench of NCWP.

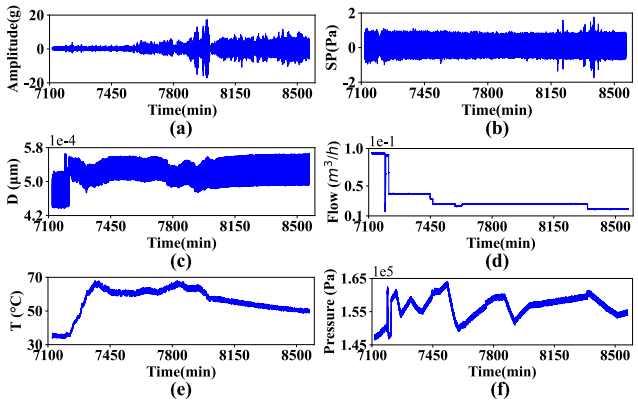



**Fig. 13.** Monitoring data of partial sensors in NCWP journal bearings from run to failure. (a) Vibration acceleration. (b) Sound pressure. (c) Shaft displacement. (d) Pipeline flow. (e) Temperature. (f) Pipeline pressure.

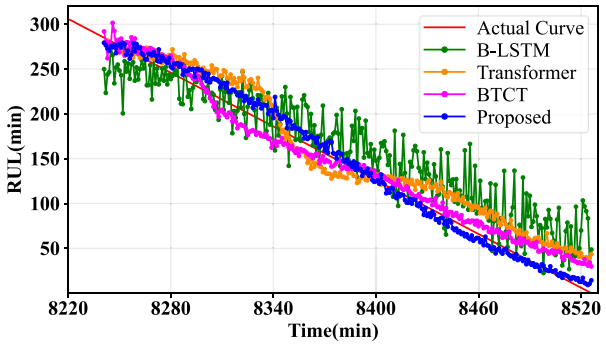



**Fig. 14.** RUL prediction results of different methods on NCWP.

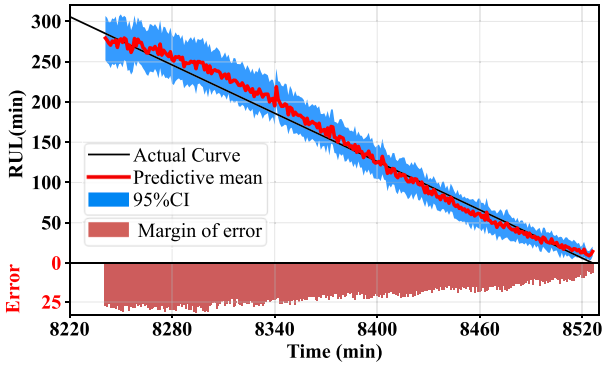



**Fig. 15.** RUL estimation results with prediction intervals on the NCWP.

### 4.2.3. RUL prediction and analysis

***(1) RUL prediction:*** As illustrated in Fig. 14, we compare the performance of the proposed method against other RUL prediction approaches on the NCWP dataset. The results indicate that the proposed method significantly outperforms the baseline methods in prediction accuracy. This improvement is primarily attributed to the following factors: (1) The UGMIF strategy enables more comprehensive and accurate extraction of degradation information. (2) The HU-model enhances prediction reliability for high-uncertainty samples. (3) The LU-model emphasizes sensor inputs with low uncertainty. (4) Interactive fusion of the HU-model and LU-model yields more accurate RUL predictions. In addition, Table 8 summarizes the quantified prediction uncertainty for various methods. The score metric used is defined in [3]. These results indicate that the proposed method achieves superior prediction accuracy. This further demonstrates its capability to address complex and challenging scenarios.

***(2) Uncertainty estimation:*** In addition, the results of the RUL uncertainty estimation is demonstrated in Fig. 15. The large increase in uncertainty compared to Case study 1 is mainly due to the fact that actual large and complex systems introduce more aleatoric uncertainties. Fortunately, the uncertainty in the predictions decreases over time, which facilitates giving reliable RUL predictions as the device approaches failure. In addition, the combined assessment of model cognitive uncertainty and aleatoric uncertainty improves the completeness and reliability of the RUL prediction uncertainty estimation.

Similarly, we quantify the uncertainty of the RUL prediction results at the later stages of NCWP degradation, as shown in Fig. 16. The figure demonstrates that the PDF of the predicted RUL covers the true RUL, and the prediction uncertainty decreases as the degradation

**Table 8**
Performance comparison of different prediction methods.

| Index | B-LSTM | B-Transformer | Transformer | BDL-based | BDL | BMNN | MAFN-QR | BTCT | TGP | HSAT | BAPT | Proposed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RMSE | 0.562 | 0.526 | 0.535 | 0.517 | 0.495 | 0.435 | 0.410 | 0.427 | 0.402 | 0.355 | 0.381 | **0.315** |
| Score | 0.295 | 0.312 | 0.305 | 0.337 | 0.354 | 0.383 | 0.411 | 0.395 | 0.407 | 0.431 | 0.425 | **0.472** |

The bold entities represent the best performance.

**Table 9**
Uncertainty estimation of different RUL prediction methods.

| $\beta$ | Metrics | B-LSTM | B-Transformer | BDL-based | BDL | BMNN | MAFN-QR | BTCT | TGP | HSAT | BAPT | Proposed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.75 | PICP | 0.711 | 0.697 | 0.675 | 0.631 | 0.694 | 0.702 | 0.703 | 0.711 | 0.731 | 0.722 | 0.745 |
| | PINAW | 0.689 | 0.565 | 0.602 | 0.649 | 0.597 | 0.561 | 0.574 | 0.537 | 0.493 | 0.518 | 0.461 |
| 0.9 | PICP | 0.859 | 0.892 | 0.846 | 0.817 | 0.839 | 0.865 | 0.871 | 0.878 | 0.905 | 0.894 | 0.913 |
| | PINAW | 0.632 | 0.457 | 0.577 | 0.642 | 0.513 | 0.506 | 0.495 | 0.501 | 0.452 | 0.487 | 0.412 |
| 0.95 | PICP | 0.897 | 0.943 | 0.917 | 0.901 | 0.931 | 0.938 | 0.942 | 0.939 | 0.959 | 0.943 | 0.954 |
| | PINAW | 0.495 | 0.431 | 0.475 | 0.514 | 0.479 | 0.456 | 0.421 | 0.427 | 0.394 | 0.405 | 0.375 |

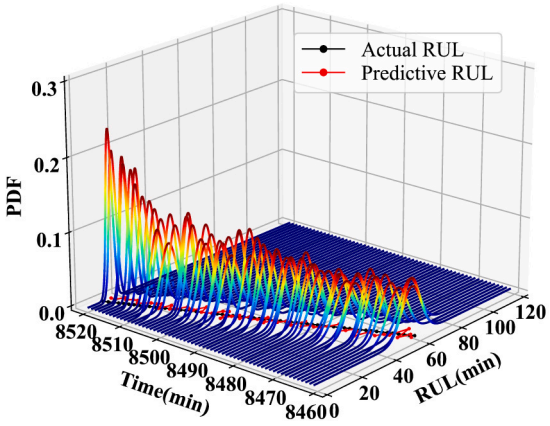



**Fig. 16.** Evaluate PDFs of RUL in journal bearing of NCWP.

progresses (higher and narrower PDF curves). This result demonstrates that the proposed method provides good prediction accuracy with lower prediction uncertainty.

The uncertainty of RUL prediction is evaluated at a confidence level denoted as $\beta = \{0.75, 0.9, 0.95\}$, as shown in Table 9. Overall, with increasing confidence levels, the uncertainty associated with all methods increases, while the reliability of the prediction intervals decreases. For instance, under a confidence level of $\beta = 0.9$, the proposed method provides more accurate uncertainty estimation. This is primarily because the proposed UGMIF strategy effectively integrates diverse monitoring signals for the complex NCWP system, thereby enhancing prediction robustness. As the confidence level increases, the uncertainty estimates produced by the proposed method become more accurate and stable, further demonstrating its superior accuracy and robustness.

***(3) Ablation experiments and analysis:*** To further validate the effectiveness of the proposed uncertainty estimation method and multi-source sensor information fusion strategy, we further carried out relevant ablation experiments as shown in Table 10. From the table it can be seen that the accuracy and uncertainty of the RUL prediction based on a single signal is still the worst. Meanwhile, without UGMIF, the prediction accuracy of network decreases while the prediction uncertainty increases, which further confirms that the fusion of multi-source information effectively improves the prediction accuracy and stability. In addition, the proposed method aims to guide the effective fusion of multi-source sensor information through uncertainty, and any source of uncertainty that is ignored results in inadequate fusion of multi-source information, reducing the RUL prediction accuracy.

***(4) UGMIF effectiveness analysis:*** To verify the validity of the LU-model, HU-model and the multi-source information fusion, the results

**Table 10**
Results of ablation experiments in NCWP.

| Metric | Single signal | w/o-UGMIF | w/o-AtU | w/o-CU | w/o-ArU | Proposed |
|---|---|---|---|---|---|---|
| RMSE | 0.601 | 0.522 | 0.491 | 0.508 | 0.475 | 0.315 |
| PICP | 0.896 | 0.916 | 0.933 | 0.925 | 0.931 | 0.954 |

w/o-UGMIF: without uncertainty-guided multi-source information fusion. w/o-AtU: without attentional uncertainty.
w/o-CU: without cognitive uncertainty. w/o-ArU: without aleatoric uncertainty. PICP is under 95%.

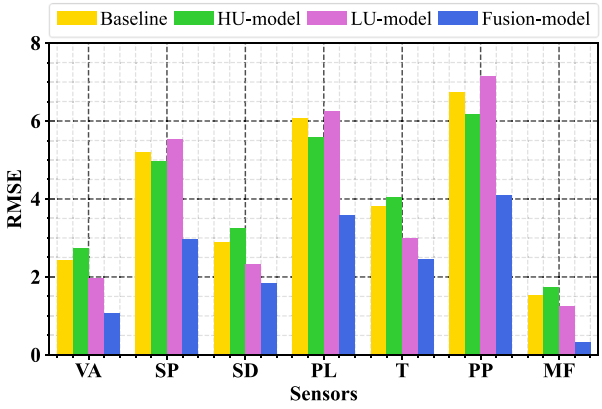



**Fig. 17.** RUL prediction performance of different prediction methods under different sensors. VA: Vibration acceleration. SP: Sound pressure. SD: Shaft displacement. PL: Pipeline flow. T: Temperature. PP: Pipeline pressure. MF: Multi-source information fusion.

are presented in Fig. 17. We find that sensor information with lower uncertainty has higher prediction accuracy. Typically, the higher uncertainty means that the model is not sensitive to information about the degradation of this type of sensor, leading to a lack of confidence in the RUL prediction. Fortunately, the proposed HU-model significantly improves the mining of sensor information with high uncertainty, such as sound pressure, pipeline flow, and pipeline pressure. The UGMIF allows these sensors to be assigned greater weights during the training process, which allows sensor information with higher uncertainty to receive more attention to improve the integrated learning of degraded features. In addition, the prediction performance is greatly improved by using the fusion of multi-source sensor information compared to single sensor RUL prediction. This result demonstrates the effectiveness and advantages of the proposed UGMIF. A novel research viewpoint for efficient fusion of multi-source sensor information is provided by the proposed UGMIF.

Similarly, we validate the prediction performance of different weighting strategies on the NCWP dataset, with the results presented

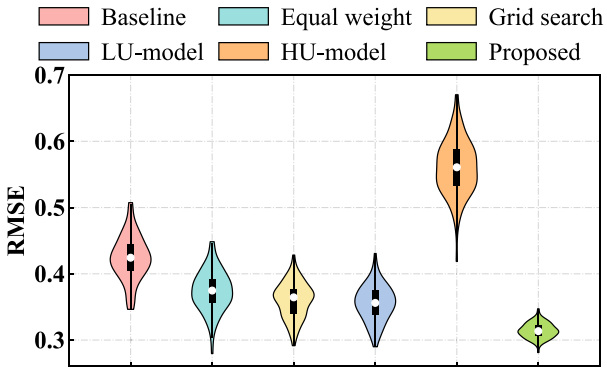



**Fig. 18.** Performance comparison of different multi-source information fusion strategies on the NCWP dataset.

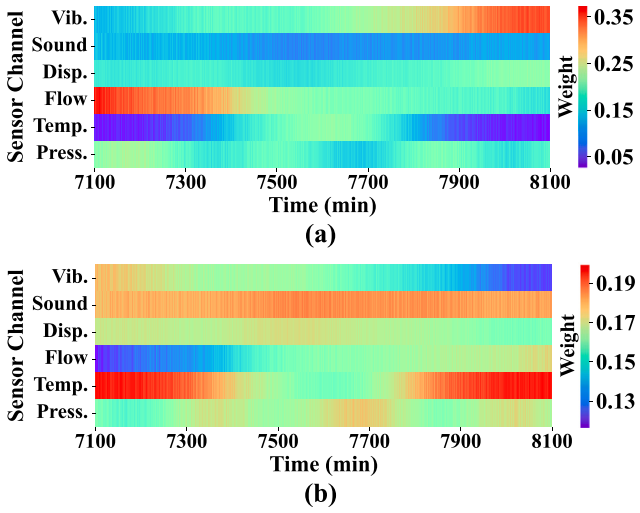



**Fig. 19.** Variation of sensor channel weights during training. (a) LU-model. (b) HU-model.

in Fig. 18. It shows that the proposed UGMIF strategy continues to outperform other methods. This advantage is primarily due to its multi-channel data fusion design, which captures both strong and weak degradation features via the LU-model and HU-model. By fully leveraging degradation feature information in different sensor channels, the method achieves accurate RUL prediction and reliable uncertainty estimation.

The dynamic changes in sensor channel weights over the course of training under the proposed UGMIF strategy in Fig. 19. In Fig. 19(a), the weights of vibration, displacement, and flow channels in the LU-model gradually increase as training progresses. This trend closely matches the patterns observed in Fig. 13 during the degradation process, indicating that UGMIF effectively captures the evolution of key sensor information to enable accurate modeling and fusion of critical features. In contrast, the HU-model in Fig. 19(b) exhibits a more balanced distribution of channel weights with less pronounced dominance, suggesting a more conservative feature fusion strategy under high uncertainty. These findings demonstrate that UGMIF can dynamically adjust the weights of different sensor channels. The LU-model assigns higher weights to key channels with high confidence, while the HU-model emphasizes channels with lower confidence and adopts a smoother, more decentralized fusion strategy. This further confirms the effectiveness and suitability of UGMIF in achieving reliable feature fusion and improving RUL prediction accuracy.

***(5) Edge deployment analysis:*** Similarly, the edge deployment performance of the proposed method is validated using the scaled NCWP dataset, and the results are presented in Table 11. Considering factors such as prediction accuracy, uncertainty, and inference latency, the optimal sample size for MC-dropout is determined to be 20. Although using a larger number of samples results in higher prediction accuracy and lower uncertainty, it also increases inference latency. Conversely, fewer samples accelerate inference but compromise prediction accuracy and reliability. Therefore, the prediction model should comprehensively consider hardware performance, prediction accuracy, uncertainty, and inference latency of edge-deployed devices. This ensures determining a reasonable number of MC-dropout samples for practical industrial deployment.

**Table 11**
The effect of MC-dropout sample size on model inference performance for the NCWP dataset.

| Metrics | MC-dropout sample size | | | | | |
|---|---|---|---|---|---|---|
| | 5 | 10 | 20 | 30 | 50 | 80 |
| RMSE | 0.317 | 0.316 | 0.315 | 0.314 | 0.314 | 0.314 |
| PICP | 0.941 | 0.943 | 0.954 | 0.954 | 0.953 | 0.953 |
| Latency (s) | 0.43 | 0.86 | 1.72 | 2.58 | 4.30 | 6.88 |

Latency refers to the mean inference delay across the test set.
The PICP is under 95%.

## 5. Conclusion

In this article, we present the Bayesian cooperative probabilistic Transformer method for mechanical RUL prediction. We model prediction uncertainty as attentional uncertainty, cognitive uncertainty, and aleatoric uncertainty to comprehensively and accurately assess RUL prediction uncertainty. Then, a new strategy for prediction uncertainty-guided multi-source information fusion is designed to improve the accuracy and robustness of model prediction by dynamically interacting and fusing the LU-model and the HU-model to jointly execute the final RUL prediction. The validity and superiority of our method are verified on the C-MAPSS dataset and the NCWP dataset. Especially, our method still has higher accuracy and robustness in the case of insufficient training data.

However, the proposed method mainly focuses on the study of RUL prediction and uncertainty estimation, and it does not realize the integration with maintenance decisions. In future work, we will further investigate how RUL with uncertainty estimation can be used to guide the customization of maintenance decision-making strategies, enabling a transition from a preventive maintenance model to a predictive maintenance model.

## CRediT authorship contribution statement

**Shushuai Xie:** Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Wei Cheng:** Supervision, Resources, Project administration. **Zelin Nie:** Resources, Project administration. **Ji Xing:** Resources, Project administration. **Xuefeng Chen:** Resources, Project administration. **Qian Huang:** Resources. **Rongyong Zhang:** Resources. **Yunjie Yang:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

The authors do not have permission to share data.

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